

Hybrid orchestration complexity:

Integrating quantum subroutines into classical workflows requires robust middleware, cross-stack compatibility, and new programming paradigms.

Technological challenges: These include limited qubit connectivity, too low gate fidelities, or large amounts of qubits required for error correction. Therefore, detecting, controlling and correcting errors becomes a major challenge.

Benchmarking and

standardisation gap: The absence of industry-wide metrics and transparent performance benchmarks hampers objective comparison of hardware and software stacks.

Vendor lock-in: Proprietary SDKs and cloud APIs lack unified standards, complicating multi-platform development and collaboration.

Collaboration woes: Lack of collaboration and exchange between industry and academia.

Cryogenic infrastructure

challenges: Quantum computers operate at temperatures close to absolute zero. Maintaining such a low temperature is a big challenge.

However, quantum computing is poised to revolutionise the healthcare, pharma, banking, logistics and materials science industries by unlocking solutions to challenges once deemed intractable.

Modern industries are actively exploring cloud-based quantum computing solutions to enhance productivity and operational efficiency. Organisations are also exploring quantum hardware roadmaps, error mitigation techniques, and hybrid quantum-classical algorithm frameworks that bridge the gap between classical systems and emerging quantum processors. **END** 

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Key quantum machine learning algorithms

Algorithm	Purpose	Applications
Quantum SVM	Fast classification by evaluating multiple possibilities simultaneously	Fraud detection, medical diagnosis
VQE (Variational Quantum Eigensolver)	Finds optimal solutions for complex optimisation problems	Molecular behaviour, chemistry simulations
QAOA (Quantum Approximate Optimization Algorithm)	Solves large scale optimisation tasks	Routing, scheduling, logistics planning

Opportunities on the horizon

The upcoming years will see open source platforms serving as the primary testing ground for developers to test new ideas in QML. India has the potential to establish itself as a top research and development hub for QML. The combination of experienced software developers with expanding quantum research funding will help India establish itself as the leading centre for QML development.

The advancement of quantum machine learning also depends on global partnerships. No single country or company can dominate this field

alone. Open source technology serves as the essential link which will transform theoretical QML into operational applications that will drive growth. **END** 

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